

Question	Answer	Marks
1(b)	Correct calculation of the mean titre • Candidate must take the average of two (or more) titres that are within a total spread of not more than 0.20 cm³. • Working/explanation must be shown OR ticks must be put next to the two (or more) accurate readings selected. • The mean should be quoted to 2 d.p. and be rounded to nearest 0.01 cm³ (e.g. 26.675 cm³ must be rounded to 26.68 cm³).	1
1(c)(i)	Final answers for (c)(ii), (c)(iii) and (c)(iv) to 3 or 4 significant figures.	1
1(c)(ii)	Correctly calculates amount of manganate(VII) = $0.01 \times \text{volume in (b)} / 1000 \text{ mol}$	1
1(c)(iii)	Correctly uses amount of Fe ²⁺ reacting with FB 2 = (c)(ii) × 5 mol AND concentration of Fe ²⁺ = ((c)(ii) × 5) × 40 mol dm ⁻³	1
1(c)(iv)	Correctly uses concentration of Fe ²⁺ = (c)(iii) × 55.8 g dm ⁻³	1
1(c)(v)	Conclusion yes / correct (because) AND (c)(iv) × 1000 / 14 (> 150 claimed as present in tablet)	1
1(d)	no because chloride (ions) are oxidised by manganate(VII) / KMnO₄. OR no because hydrochloric acid is monoprotic AND might not produce enough H⁺. OR yes because hydrochloric acid is a strong acid.	1